Lesson plan · for the teacher

Pipelines, quality and governance: unit check

Data engineering · Level 4 · Unit check

Time	Materials
40 min	Board, chalk, paper. No devices needed.

Learning goals

- Automate the weekly report
- Checks that catch problems
- Who may see what
- Keep, archive, delete

Activity: Team quiz

1. Teams (2 min): Make teams of four and give each team a sheet of paper for its answers.
2. Quiz (20 min): Read each question aloud from the quiz sheet; each team agrees on one answer and writes it down.
3. Explain (10 min): Reveal each answer and ask one team to explain why it's right, using an example from the unit.
4. Wrap-up (8 min): Write the question most teams got wrong on the board, and work through it together.

Discussion

- Which report would you automate first?
- Who sees too much in our school today?
- When did we last test a restore?

Exit ticket

- One change I will propose to our head this term, and why.

Try it in class tomorrow

Pick one report your school sends, and write down its pipeline, its checks, who may see it and how long its data is kept: one line each.

Worksheet · for pupils

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Name _____ Class _____ Date _____

1 **Put a pipeline's steps in order.**

- [ ] Log the run
- [ ] Load
- [ ] Transform
- [ ] Extract

2 **A load fails halfway and the pipeline reruns. Which load gives the right report?**

◯ One that replaces the day's rows ◯ One that adds the rows again

◯ Either: the computer sorts it out

3 **Retrying fixes a register that says "?" instead of P or A.**

◯ True ◯ False

4 **Match each check to the SQL that runs it.**

In range	• • • •	GROUP BY station, day HAVING COUNT(*) > 1
Not empty	• • • •	WHERE mm IS NULL
No repeats	• • • •	WHERE mm < 0 OR mm > 300

5 **This check never finds the missing marks. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  SELECT pupil_id
2  FROM marks
3  WHERE mark = NULL;
```